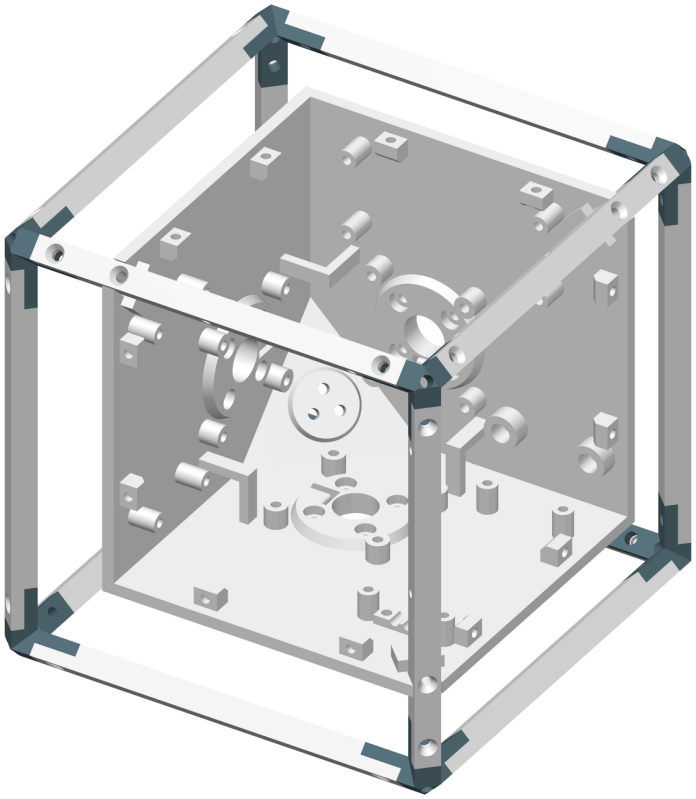
Heat-set inserts

5 steps, about 1 minute per insert



Designed by Athanasia Nikolova

v1.1 · October 2026

All data come from tests and many real builds, not only from theory.

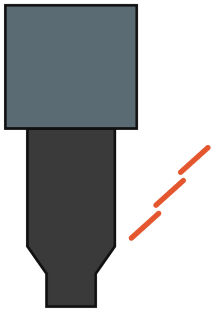
Teacher only

- **One thing per page**
  One action, one big picture. The part of the step is the only strong colour, the rest is quiet grey. Short sentences, much white space.
- **Colour is never the only clue**
  Every wire on a board drawing also has a number, every board a letter (X, Y, Z), every insert its size. The insert colours stay distinct with colour-vision deficiency (Okabe-Ito palette).
- **Readable**
  All instruction text has a contrast of at least 4.5 : 1 to the page (WCAG 2.x, AA).
- **Print it right**
  In colour, at 100 % - not 'fit to page'. The Assembly guide draws every screw at real size: the student lays the screw on the page to check it.
- **Pace**
  One page, then the action, then the next page. For a student who tires quickly: one chapter per session. The check pages are good stopping points.

✋ Safety first

- **Grown-up nearby**
  Work with your teacher or mentor in the room.
- **Hot!**
  The tip is 220 °C. Put it back in the stand every time.
- **Fresh air**
  Melted plastic smells. Open a window.
- **Unplug**
  When you stop, switch the iron off and let it cool.

You need



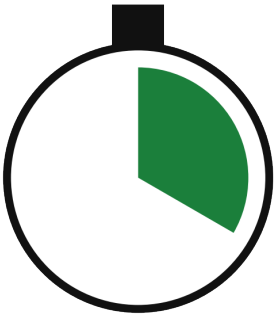
Soldering iron



Brass inserts



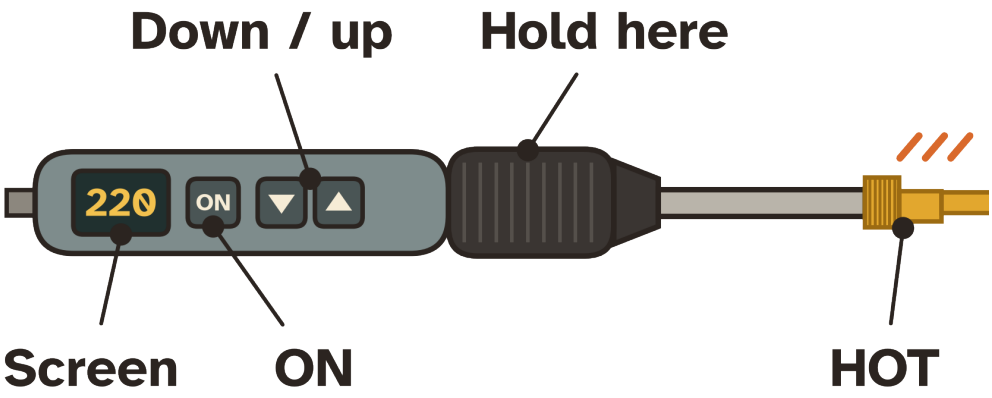
Flat metal block



10 seconds

Your iron

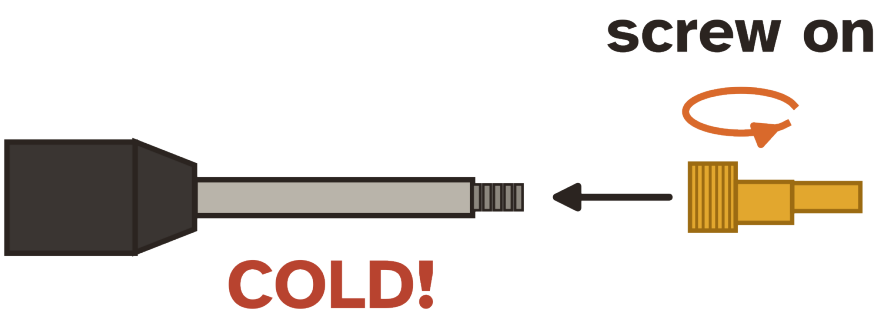
USB-C soldering iron



100–450 °C · 100 W · USB-C · Hot in seconds

Hold the black grip. Never the metal.

Put on the insert tip

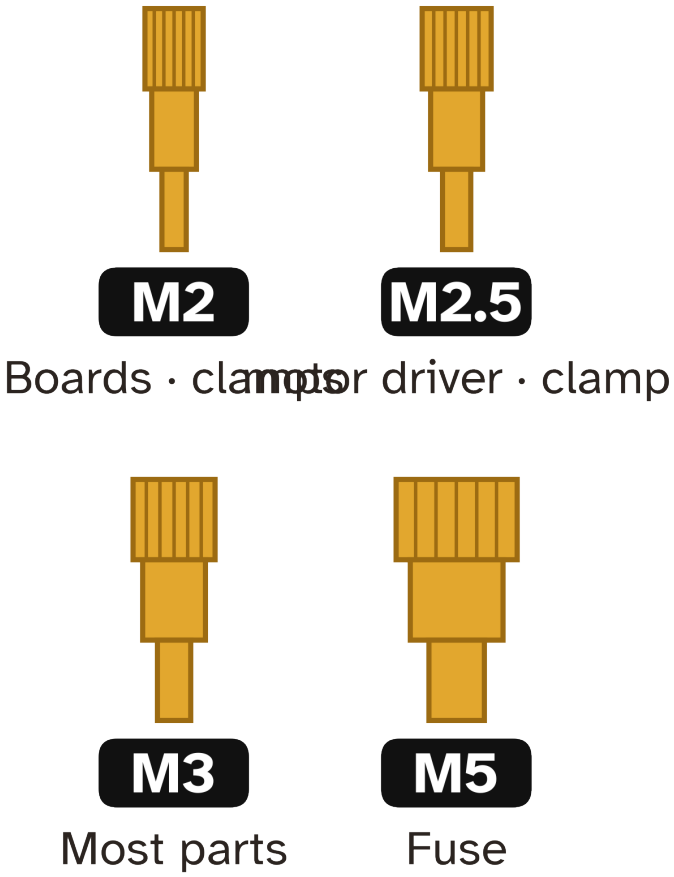


Iron OFF. Tip cold.

Screw the brass tip on by hand.

Which tip?

Same size as the insert.



Change the tip only when it is cold.

Know your inserts

Each kind has its own colour.



M3

SHORT 4 mm

×91



M3

SHORT 3 mm

×3



M3

NORMAL 5.7 mm

×14



M2.5

4 mm

×14



M2

3 mm

×12



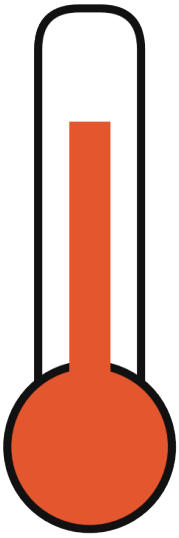
M5

5.8 mm

×2

Drawn at the same scale. All M3 use the same hole: Ø4.0.

Set the iron



220 °C

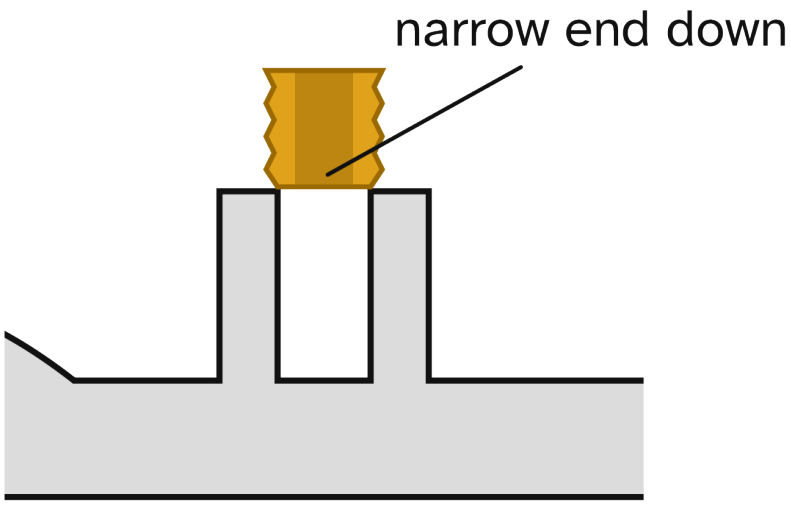
PETG-CF

Up / down to 220.

Wait until it is hot.

1 Place

Cut through a real motor driver standoff (box, M2.5)

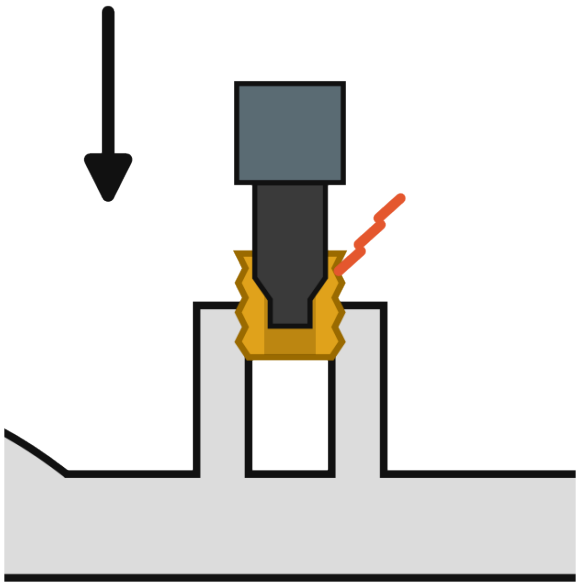


Narrow end down.

Straight on the hole.

2 Press

Cut through a real motor driver standoff (box, M2.5)

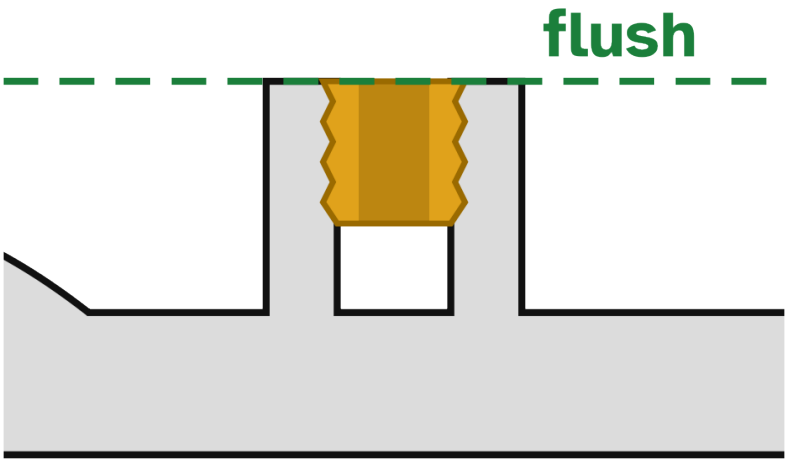


Slow and straight.

Let the heat do the work.

3 Stop when level

Cut through a real motor driver standoff (box, M2.5)

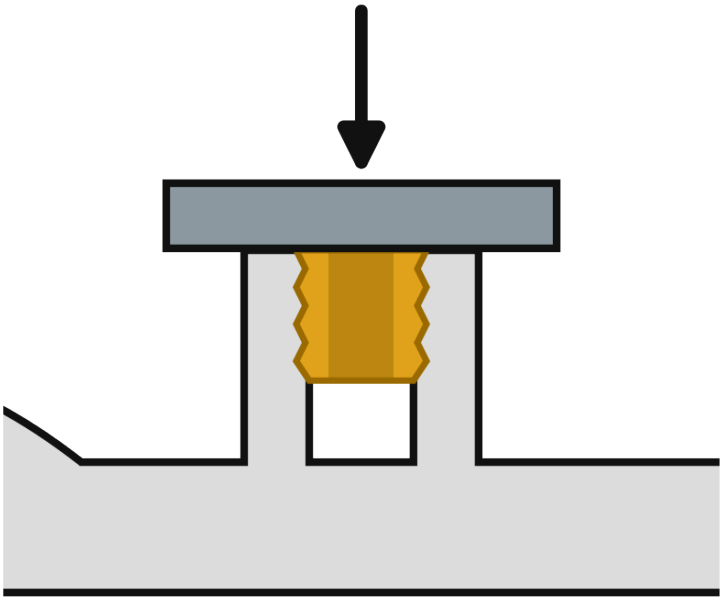


Top of the insert =

top of the part.

4 Hold flat

Cut through a real motor driver standoff (box, M2.5)

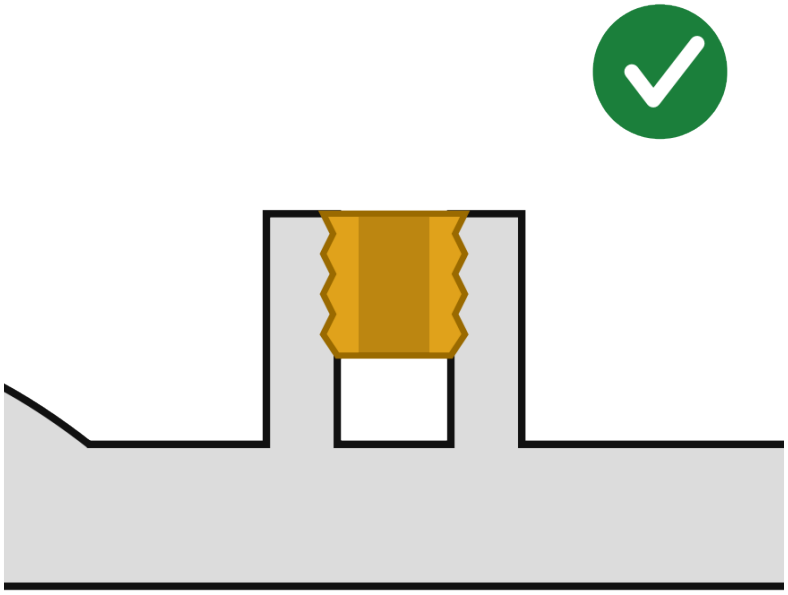


Press the metal block on top.

Count to 10.

5 Check

Cut through a real motor driver standoff (box, M2.5)

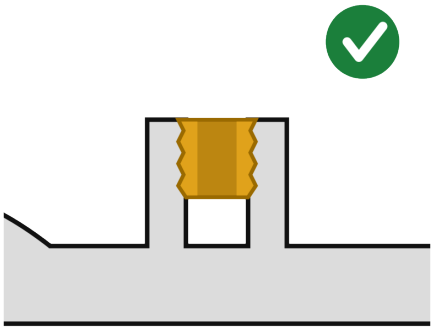


Straight? Level?

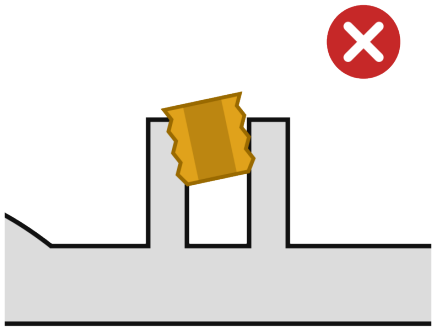
Done. Next one.

Good or not?

Cut through a real motor driver standoff (box, M2.5)

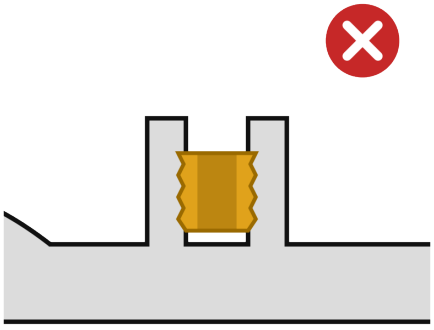


Good



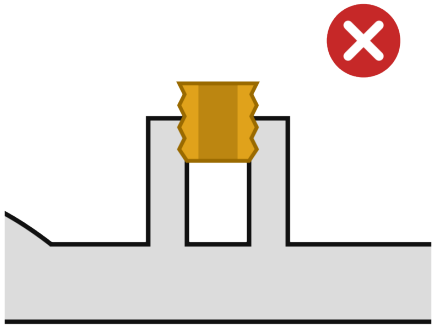
Tilted

Reheat. Straighten.



Too deep

Screw must still reach.



Sticks out

Reheat. Press flat.

Where they go

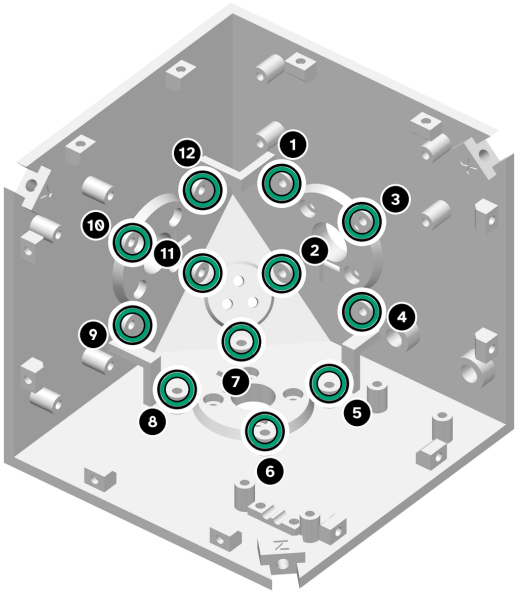
Tick each place when it is done.

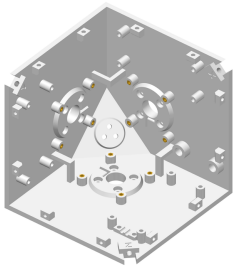
	place		page
☐	Box inside: motor driver	● ×12	15
☐	Box inside: boards	● ×12	16
☐	Box inside: fuse	● ×2	17
☐	Box inside: clamp	● ×2	18
☐	Box outside: corner 0	● ×3	19
☐	Box outside: ties	● ×3	20
☐	Tube: cone end	● ×3	21
☐	Tube: IMU pocket	● ×4	22
☐	Tube: table top	● ×4	23
☐	Basin: top plate	● ×3	24
☐	Corners: arms	● ×3 on 8 parts	25
☐	Corners: tip	● ×1 on 8 parts	26
☐	2 corners: inside	● ×1 on 2 parts	27
☐	Navels: legs	● ×6 on 8 parts	28
☐	Lower arcs: middle	● ×1 on 6 parts	29

Box · inside

Motor faces: 4 standoffs on each

M2.5 4 mm **×12**





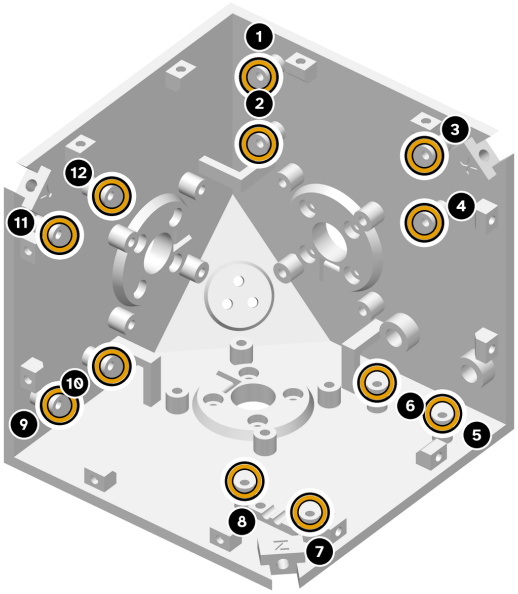
Done: like this

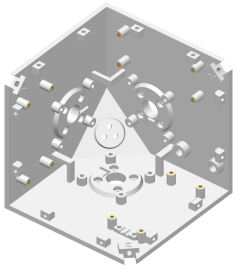
Later a screw goes in here: motor driver standoffs.

Box · inside

Board standoffs: 4 under each board

M2 3 mm **×12**





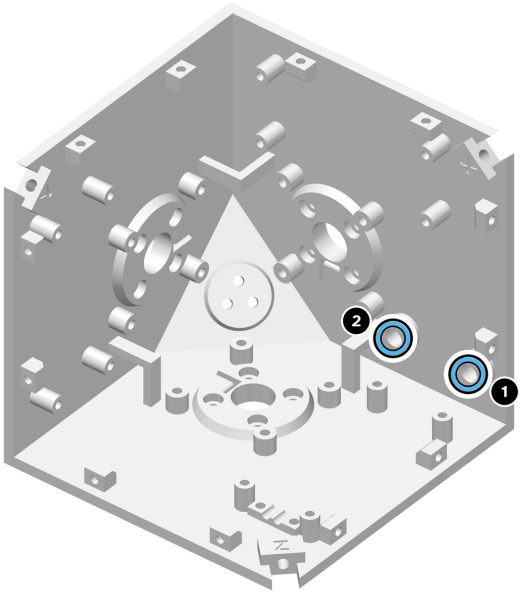
Done: like this

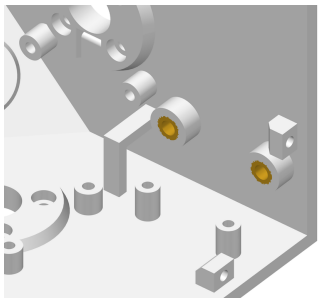
Later a screw goes in here: board standoffs.

Box · inside

The 2 fuse bosses, face x

M5 5.8 mm **×2**





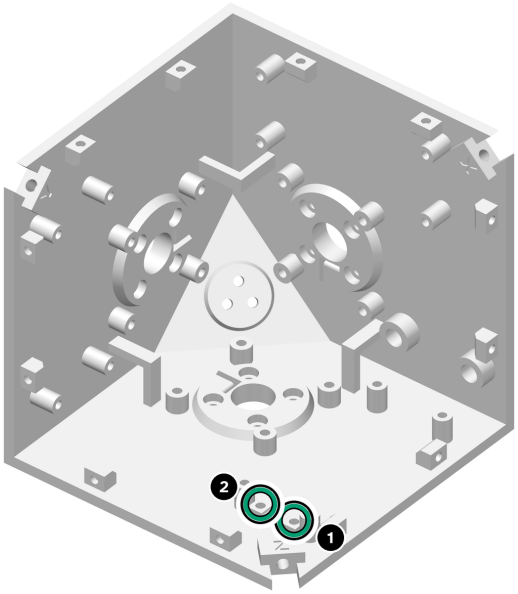
Done: like this

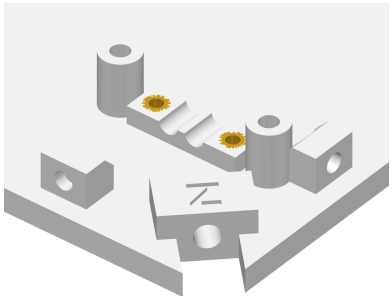
Later a screw goes in here: MIDI fuse terminals.

Box · inside

Cable clamp base, face z

M2.5 4 mm **×2**





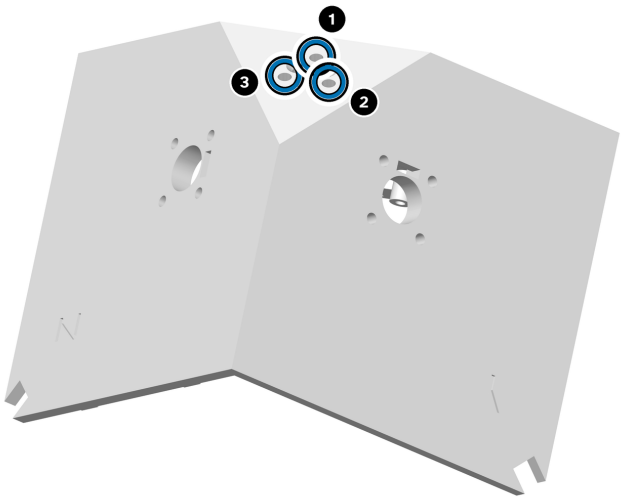
Done: like this

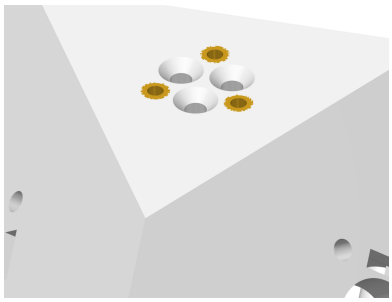
Later a screw goes in here: cable clamp bar.

Box · outside

Turn the box over: corner 0 up

- **M3 SHORT 4 mm** **×3**





Done: like this

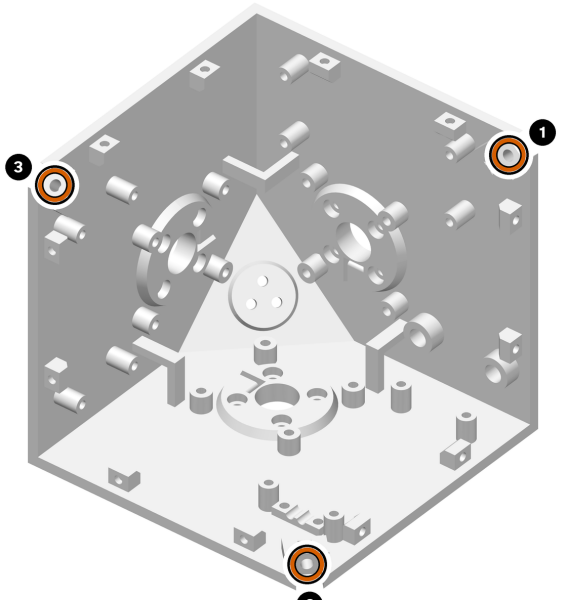
Later a screw goes in here: box standoff flange.

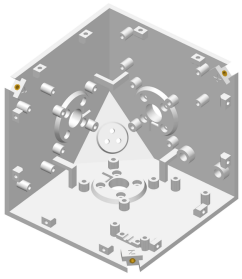
Here goes the box standoff flange.

Box · outside

The 3 cage ties, on the edges

M3 NORMAL 5.7 mm **×3**





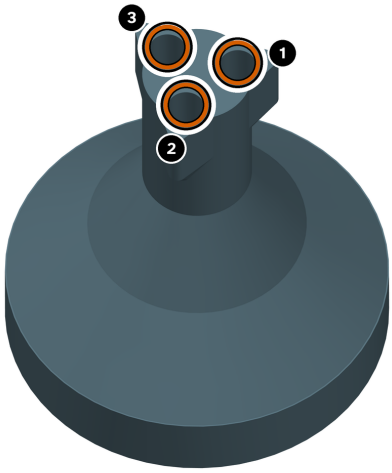
Done: like this

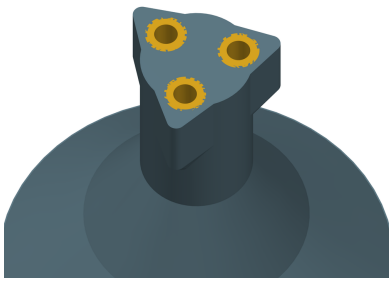
Later a screw goes in here: cage to Cubi.

Battery tube

Cone end up - as it comes off the printer

M3 NORMAL 5.7 mm **×3**





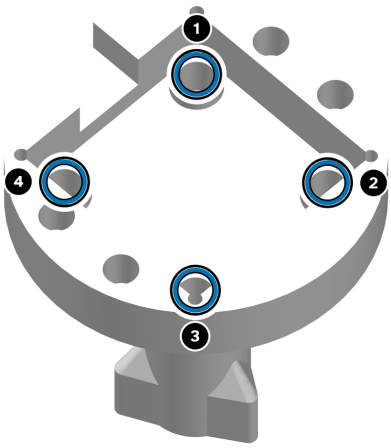
Done: like this

Later a screw goes in here: from outside the box.

Battery tube

Turn it over: the table up - the 4 in the pocket

M3 SHORT 4 mm **×4**





Done: like this

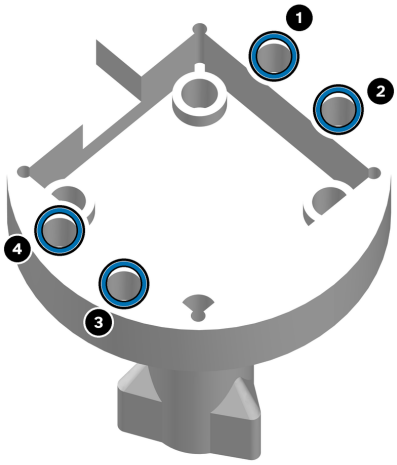
Later a screw goes in here: IMU screws.

The IMU board sits in this pocket, on its 4 bosses.

Battery tube

The table up - the 4 on the table top

- **M3 SHORT 4 mm** **×4**





Done: like this

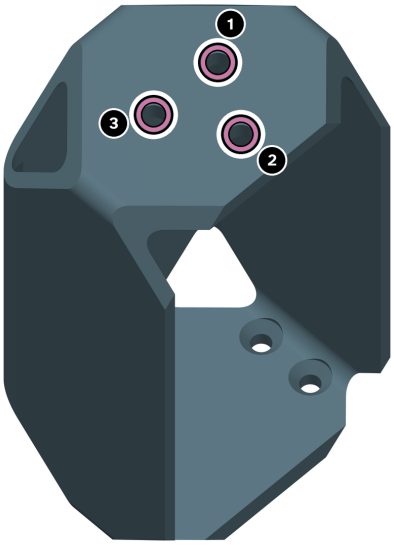
Later a screw goes in here: basin screws.

The basin is screwed here, over the IMU.

Battery basin

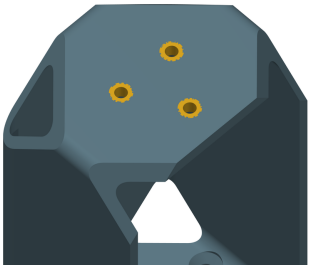
Top plate, from the outside

M3 SHORT 3 mm **×3**



Done: like this

Later a screw goes in here: hub standoff flange.

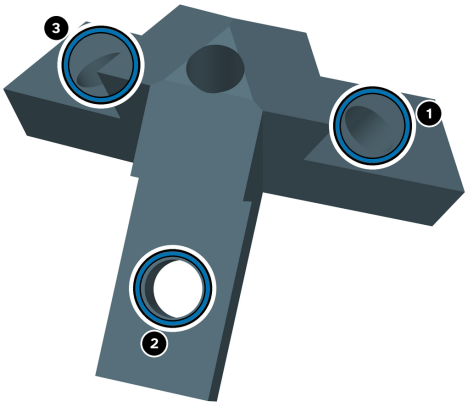


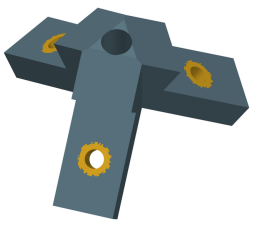
Cage corner · all 8

The 3 arm ends, from the outside

M3 SHORT 4 mm **×3**

on each part · 8 parts = 24 inserts





Done: like this

Later a screw goes in here: cage beams.

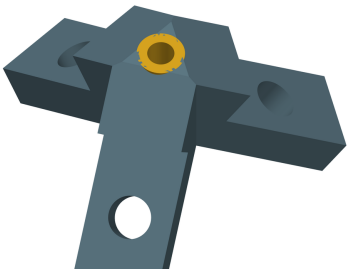
Cage corner · all 8

The tip, from the outside

M3 NORMAL 5.7 mm **×1**

on each part · 8 parts = 8 inserts





Done: like this

Later a screw goes in here: navel.

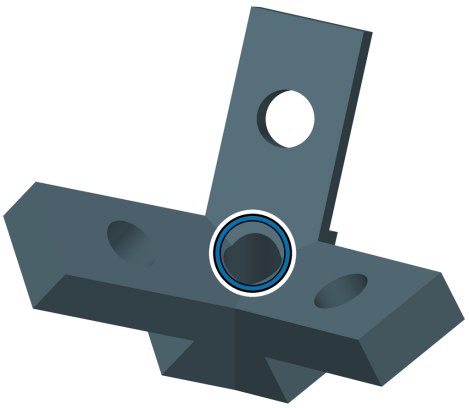
2 corners get one more insert, inside - next page.

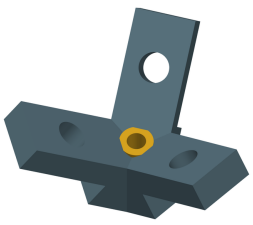
Cage corner · only 2

The tip, from the INSIDE

M3 SHORT 4 mm **×1**

on each part · 2 parts = 2 inserts





Done: like this

Later a screw goes in here: box standoff / hub standoff.

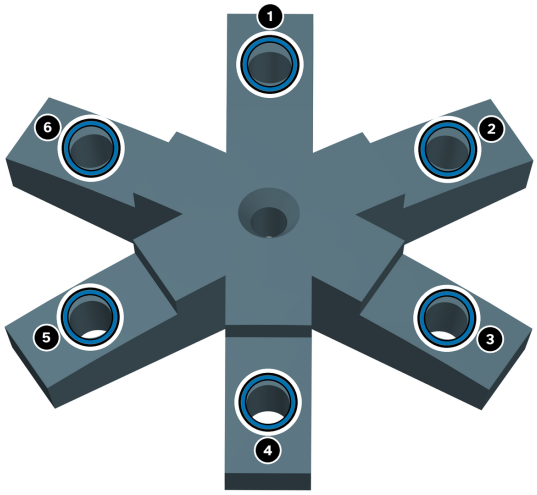
Pick 2 corners, mark them with a pencil: corner 0 and corner (1,1,1). Same hole as the tip insert.

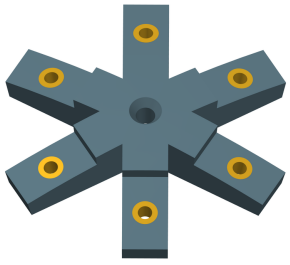
Navel · all 8

The 6 legs, from the top

M3 SHORT 4 mm **×6**

on each part · 8 parts = 48 inserts





Done: like this

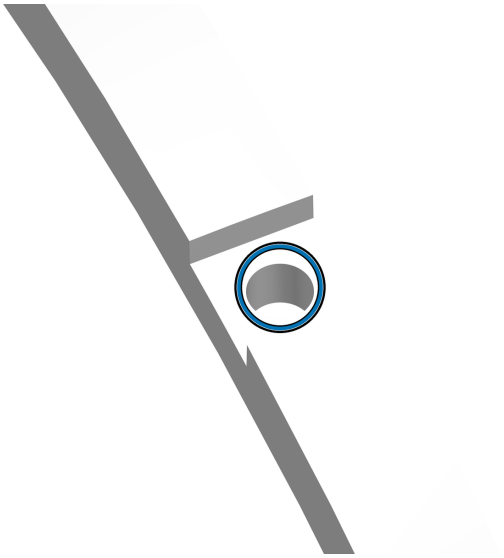
Later a screw goes in here: sphere arcs.

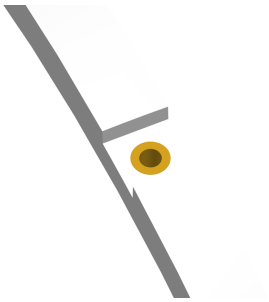
Diagonal arc · LOWER, all 6

The middle, where the arcs cross

M3 SHORT 4 mm **×1**

on each part · 6 parts = 6 inserts





Done: like this

Later a screw goes in here: upper diagonal arc.

Only the 6 LOWER arcs. The 6 upper arcs have no insert.

Good to know

- **No force**
  If it does not go in, wait. Heat, not muscle.

- **Wrong one?**
  Heat it again and pull it out with the iron.